MS Applied Data Science

THE UNIVERSITY OF CHICAGO

ADSP 31021 Machine Learning Operations
Assignment #2 – Feature Store
Due Date – Beginning of Session #5

Assignment Overview
The goal of this assignment is to provide hands-on experience building a reproducible machine learning workflow using MLOps principles. Students will design and implement an end-to-end pipeline that integrates feature management, experiment tracking, versioning, and model evaluation.

Reproducibility Requirements
- All code must be executable from a clean environment.
- Include dependency management files (requirements.txt, environment.yml, etc.).
- Provide instructions for reproducing results.
- Ensure experiment results can be reproduced using the submitted repository.
- Version datasets, features, or artifacts where appropriate.

Required Tools
- Python 3.x
- Git and GitHub
- An MLOps platform (examples: MLflow, Kubeflow, Airflow, DVC, etc.)
- A feature store framework or service
- Libraries required for model development and evaluation

Instructions

1. Dataset Selection and Setup (10 Points)
- Use the provided athletes.csv dataset.
- Perform any required exploratory analysis and preprocessing necessary for your pipeline.
- Clearly document any assumptions or modifications made to the dataset.

2. MLOps Platform Selection and Configuration (10 Points)
- Select and configure an MLOps platform of your choice.
- Clearly explain why you selected this platform and how it supports your workflow.

3. End-to-End ML Pipeline Development (20 Points)
Build an end-to-end machine learning pipeline that includes:
- Data ingestion
- Data preprocessing
- Feature engineering / feature creation

- Model training
- Model evaluation

Your pipeline should be modular, reusable, and executable from a clean environment.

4. Feature Store Integration (15 Points)

- Integrate a feature store into your machine learning workflow.
- Demonstrate how features are stored, retrieved, and utilized within the pipeline.

5. Feature Versioning (15 Points)

- Create at least two different versions of feature definitions.
- Store and manage these versions appropriately within your selected tooling.
- Clearly explain the differences between feature versions.

6. Experimentation and Model Training (20 Points)

You must run experiments using combinations of:

- Two different feature versions
- Two different hyperparameter configurations
- The same algorithm across all experiments

This should result in **four total experiment combinations.**

Requirements:

- AutoML and automated algorithm selection are NOT permitted.
- Experiment tracking should clearly document parameters, feature versions, and results.

7. Documentation and Professional Practices (10 Points)

Your submission should demonstrate:

- Clear documentation
- Reproducibility
- Proper repository organization
- Readable, maintainable code

Assignment Requirements

- Create two different feature versions.
- Train one algorithm using two hyperparameter configurations.
- Combine feature versions and hyperparameters to create four total experiments.
- Document assumptions and decisions made throughout the workflow.

Deliverables

- GitHub repository containing working code
- README with setup and execution instructions
- Experiment comparison summary
- Model evaluation metrics and visualizations
- Evidence of feature versioning and experiment tracking